

College of Science, Department of Mathematics
Bachelor of Science in Mathematics, **Applied Computational Mathematics Option**
For students entering under UG Catalog **2022-2023**

I. Pathways General Education Requirements (47 credits)

Concept 1: Discourse.

(1f) Foundational Discourse (6 credits)

ENGL 1105: First-Year Writing	3	☐
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ENGL 1106: First-Year Writing*	3	☐
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(1a) Advanced or Applied Discourse (3 credits)

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Concept 2: Critical Thinking in the Humanities (6 credits)

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Concept 3: Reasoning in the Social Sciences (6 credits)

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Concept 4: Reasoning in the Natural Sciences (6 credits)

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Concept 5: Quantitative and Computational Thinking

(5f) Foundational Quantitative and Computational Thinking (8 credits)

MATH 1225: Calculus of a Single Variable*	4	☐
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MATH 1226: Calculus of a Single Variable*	4	☐
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(5a) Advanced or Applied Quantitative and Computational Thinking (3 credits)

MATH 2214: Intro Diff Equations*	3	☐
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Concept 6: Critique and Practice in Design and the Arts (6 credits = [3 in design + 3 in arts] or [6 in integrated design and arts])

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Concept 7: Critical Analysis of Identity and Equity in the United States (3 credits)

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In Pathways, some courses can be used for Concept 7 plus one other concept, but no other double-counting is permitted.

II. Mathematics Bachelor of Science Core Courses (21 credits)

MATH 2114: Introduction to Linear Algebra*	3	☒	3 ☒
Multi-D Calculus: 6 credits from – MATH 2204 Intro Multivariable Calculus* MATH 2214 Intro Diff Equations* MATH 3214 Calculus of Several Variables* MATH 2204, 2114, and 3214 are required for ACM. MATH 2214 = Pathway 5a MATH 2204+3214 = Core	3	☒	
MATH 3034: Introduction to Proofs*	3	☒	
MATH 3144: Linear Algebra I*	3	☒	
MATH 3224: Advanced Calculus*	3	☒	
Computer Programming 3 credits from – MATH 1454 Intro to Math Programming* MATH 3054 Programming for Math* CS 1044 Intro Prog in C CS 1114 Intro to Software Design If MATH 4414 is planned in Section III, need prereq of MATH 1454 or MATH 3054	3	☒	

***Some courses listed on this checksheet may have prerequisites and/or corequisites; please consult the University Course Catalog or check with your advisor**

III. Required Courses Specific to the Applied Computational Mathematics Option (15 credits)MATH 4425: Fourier Series and Partial Differential Equations* and
(MATH 4426: Fourier Series and Partial Differential Equations* or
CMDA 4604: Intermediate Topics in Mathematical Modeling*)

6

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MATH 4445, 4446: Numerical Analysis*

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☐MATH 4414: Scientific Computing* or MATH 4454 Applied
Mathematical Modeling*

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☐**IV. Restricted Electives (6 credits)****Mathematics Electives**

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The six hours of math electives must be chosen from the Mathematics courses numbered between 4044 and 4454, with the following exceptions: (a) MATH 3124* can be used to satisfy three of the six hours. (b) No more than three hours from MATH 4044*, 4334*, 4344* can be used.

Applied Area Courses (12 Credits. Must be approved by the Undergraduate Program Committee)

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☐**V. Free Electives (Sufficient to achieve the 120 credit graduation requirement)**

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*Some courses listed on this checksheet may have prerequisites and/or corequisites; please consult the University Course Catalog or check with your advisor.

VI. Outcomes Assessment

Each student is required to participate in the department's Outcomes Assessment procedures as determined by each year's Undergraduate Program Committee and approved by the Chair.

VII. Minimum Hours Required for Graduation

120 Credits

VIII. Foreign Language Requirement

Students who did not successfully complete at least two years of a single foreign, classical, or sign language during high school must successfully complete six credit hours of a single foreign, classical, or sign language at the college level. Courses taken to meet this requirement do not count toward the hours required for graduation. Please consult the Undergraduate Catalog for details.

IX. Satisfactory Progress to Degree

Upon having attempted 36 semester credits, the student must have completed 12 credits of the Pathways General Education Requirements. Upon having attempted 72 credits (including transfer, advanced placement, advanced standing, credit by examination, and course withdrawal), the student must have completed 24 credits of the Pathways General Education Requirements. In addition, satisfactory progress toward the B.S. in mathematics requires that:

- 1) Within the previous two semesters, the student must pass at least one mathematics course that is used in the in-major GPA calculation.
- 2) Upon having attempted 45 semester credits, students must have an in-major GPA of 2.2 or above.
- 3) Upon having attempted 72 semester credits (including transfer, advanced placement, advanced standing, credit by examination, and course withdrawal), students must have completed the following courses with grades of C- or better: MATH 1225, 1226, 2114, 2204, 2214, and 3034, and not have taken any of these courses more than twice, including attempts ending in course withdrawal.

X. Minimum GPA Required for Graduation

Students are required to have a 2.0 GPA and a 2.0 in-major GPA for Graduation. In-major GPA for this option is computed using all MATH courses with the exception of MATH 1014, 1015, 1016, 1025, 1026, 1524, 1525, 1526, 1535, 1536, 1614, 1624, 2004, 2015, 2016, 2024, 2534, 2644, 3624, 4574, 4625, 4626, 4644, 4654, 4664.